

AI in Clinical Practice — Complete Course Package

Unified NIH/AAMC + German/EU Curriculum

Course dates: October 1, 2026 - March 24, 2027 (25 weeks) Format per week: 3 hrs Virtual Classroom + 3 hrs In-Hospital Practical Sandbox = 6 hrs/week (24 hrs/month)

This package contains: (1) the full week-by-week course material and schedule, (2) the reference library mapped to each week, and (3) the USMLE-style discussion question bank for each session (carried over from the prior document, now aligned to exact weeks).

MASTER CALENDAR

Month	Weeks	Dates	Chapter
1	1-4	Oct 1 - Oct 28, 2026	Admin Time Savings & AI Foundation
2	5-8	Oct 29 - Nov 25, 2026	AI-Driven Diagnostics & Safety Override
3	9-12	Nov 26 - Dec 23, 2026	Clinical Care Pathways & Data Sovereignty
4	13-17	Dec 24, 2026 - Jan 27, 2027	Ethicolegal Autonomy & Final Checkoff (5 wks, incl. holiday buffer)
5	18-21	Jan 28 - Feb 24, 2027	Consolidation & Mock OSCE
6	22-25	Feb 25 - Mar 24, 2027	Solo Pilot Capstone & Certification

MONTH 1 — ADMIN TIME SAVINGS & THE AI FOUNDATION

Focus: Eliminate charting burnout; build HITL literacy and prompt-engineering fluency.

Week 1 (Oct 1-7): Introduction to AI in Medicine

Virtual (3 hrs)

- Course orientation; landscape of clinical AI (scribes, CDSS, imaging AI, care-planning NLP)
- What is “Human-in-the-Loop” (HITL) and why it is the organizing principle of the entire course
- Foundational ML literacy: supervised vs. generative models, what a “hallucination” is

at a technical level (plausible-sounding but false output) **Hospital Practice (3 hrs)**

- Orientation to EHR workstations and ambient scribe hardware in the sandbox environment
- Baseline documentation-time measurement (pre-AI) for later comparison **Key References**
- HITL workflow overview — pubmed.ncbi.nlm.nih.gov/42710030
- NIH/AAMC Clinical AI Competency Model — pubmed.ncbi.nlm.nih.gov/36731054
Discussion: Q1.1 (automation bias sentinel case)

Week 2 (Oct 8-14): Prompt Engineering — The “Four Rs”

Virtual (3 hrs)

- AAFP Four Rs framework: Request, Reason, References, Revise
- Live-prompting workshop: write and critique prompts for note summarization, differential generation, and patient-instruction drafting
- Common failure modes: vague requests, no verification step, no source grounding **Hospital Practice (3 hrs)**
- Supervised prompting of the ambient scribe on standardized-patient encounters
- Peer review of AI-generated notes against the audio transcript **Key References**
- AAFP “Four Rs” framework — aafp.org/fpm/2025/0900/effective-ai-prompts
- Prompt literacy in clinical training — pmc.ncbi.nlm.nih.gov/articles/PMC11822726
Discussion: Q1.2 (Four Rs prompt structure)

Week 3 (Oct 15-21): Ambient Scribing in Practice

Virtual (3 hrs)

- Deep dive on ambient AI medical scribes: architecture, latency, and error taxonomy (mistranscription, omission, hallucinated detail)
- German GP digitalization gap: why structured onboarding (not self-study) is mandated **Hospital Practice (3 hrs)**
- Full mock clinic day using ambient scribes across varied patient complexity
- Students log every correction they make to the AI draft (“catch log”) **Key References**
- German GP digital literacy gap — germancomplianceinstitute.com (2026 compliance guide)
- Clinical AI evidence base — pmc.ncbi.nlm.nih.gov/articles/PMC11993958 **Discussion:** Q1.3 (German literacy driver)

Week 4 (Oct 22-28): Month 1 Synthesis & Checkoff

Virtual (3 hrs)

- Review “catch logs” as a class; quantify time saved vs. errors caught
- Formal introduction of the “efficiency ≠ trust” principle **Hospital Practice (3 hrs)**

- Month 1 practical checkoff: independently run one full ambient-scribed encounter with a signed, corrected note **Key References**
- Ambient scribing outcomes review — aafp.org/fpm/2025/0900 **Discussion:** Q1.4 (efficiency vs. trust) **Milestone:** Month 1 Competency Checkoff (Admin/Prompting) — pass/fail gate to Month 2

📅 MONTH 2 — AI-DRIVEN DIAGNOSTICS & THE SAFETY OVERRIDE

Focus: Critical appraisal of AI diagnostic output; manual override competency.

Week 5 (Oct 29–Nov 4): Critical Appraisal of AI Diagnostics

Virtual (3 hrs)

- Evidence review: AI board-exam performance vs. real-world bedside reasoning dissociation
- Anatomy of a “confident hallucination” in diagnostic AI (high stated confidence ≠ correctness) **Hospital Practice (3 hrs)**
- Shadow sessions on hospital CDSS / radiology AI platforms; observe live outputs without acting on them yet **Key References**
- AI board-exam vs. bedside reasoning dissociation — news.syenza.com/ai-models-german-medical-licensing-exams
- Clinical AI evaluation methods — pmc.ncbi.nlm.nih.gov/articles/PMC12820482 **Discussion:** Q2.4 (board-score dissociation)

Week 6 (Nov 5–11): German Quality Pillars — Functionality, Robustness, Comprehensibility

Virtual (3 hrs)

- Deep dive into the three pillars and how each maps to a distinct failure mode
- Explainable AI (XAI) under GDPR: what “the right to explanation” requires in practice **Hospital Practice (3 hrs)**
- Case review: identify which pillar was violated in real de-identified CDSS malfunction reports **Key References**
- German quality pillars — physicianaihandbook.com/future/medical-education.html
- GDPR/XAI compliance — pmc.ncbi.nlm.nih.gov/articles/PMC12978887 **Discussion:** Q2.2 (Comprehensibility pillar violation)

Week 7 (Nov 12–18): The Sandbox Override Exercise

Virtual (3 hrs)

- Debrief methodology: how instructors seed anomalous/biased data and what “successful override” looks like

- Bias in training data: demographic underrepresentation and its downstream risk-score effects **Hospital Practice (3 hrs)**
- Live sandbox exercise: students work a shift where the CDSS is deliberately flawed; graded on detection + correct manual override + documentation **Key References**
- Clinical AI bias case studies — pmc.ncbi.nlm.nih.gov/articles/PMC11993958
Discussion: Q2.1, Q2.3 (override scenarios)

Week 8 (Nov 19-25): Month 2 Synthesis & Checkoff

Virtual (3 hrs)

- Class-wide debrief of override exercise outcomes; discuss near-misses **Hospital Practice (3 hrs)**
- Month 2 practical checkoff: independently detect and override a seeded CDSS error under observation **Milestone:** Month 2 Competency Checkoff (Diagnostics/Override) — pass/fail gate to Month 3

🏠 MONTH 3 — CLINICAL CARE PATHWAYS & DATA SOVEREIGNTY

Focus: AI-assisted care planning; GDPR compliance and local/open-weight deployment rationale.

Week 9 (Nov 26-Dec 2): AI-Enhanced Care Planning

Virtual (3 hrs)

- NLP-extracted problem lists combined with standardized nursing care models (e.g., NANDA-I)
- Where HITL applies in nursing workflows specifically **Hospital Practice (3 hrs)**
- Nurses/students draft AI-assisted care plans and revise against actual bedside presentation **Key References**
- AI-enhanced clinical encounters — pmc.ncbi.nlm.nih.gov/articles/PMC7980122
Discussion: Q3.2 (nursing HITL review)

Week 10 (Dec 3-9): GDPR & Data Sovereignty

Virtual (3 hrs)

- Why EU law frequently blocks cross-border cloud API use for identifiable health data
- Case study: a hospital blocked from using a non-EU cloud LLM for care-plan generation **Hospital Practice (3 hrs)**
- Walkthrough of the hospital's actual data-flow diagram; identify every point where patient data would leave institutional control **Key References**
- EU GDPR healthcare compliance guide — [germancomplianceinstitute.com](https://www.germancomplianceinstitute.com)

- Local/open-weight deployment rationale — vde.com/de/dgbmt event proceedings
Discussion: Q3.1, Q3.4 (cross-border transfer, open-weight rationale)

Week 11 (Dec 10-16): Anonymization & Secondary Data Use

Virtual (3 hrs)

- Anonymization vs. pseudonymization; data-minimization principles for research use
- The DiGA (Digital Health Applications) reimbursement pathway as a compliance model
Hospital Practice (3 hrs)
- Practice extracting an anonymized dataset from AI-generated documentation for a mock research request **Key References**
- DiGA pathway — pmc.ncbi.nlm.nih.gov/articles/PMC12973773 **Discussion:** Q3.3 (research data compliance)

Week 12 (Dec 17-23): Month 3 Synthesis & Checkoff

Virtual (3 hrs)

- Review local EHR-embedded AI modules (server-hosted) as the German-preferred deployment model **Hospital Practice (3 hrs)**
- Month 3 practical checkoff: produce one GDPR-compliant, anonymized care-plan data extract **Milestone:** Month 3 Competency Checkoff (Care Pathways/Sovereignty) — pass/fail gate to Month 4 (*Winter break recommended after Week 12 if the institution observes a holiday recess before Dec 24.*)

■ MONTH 4 — ETHICOLEGAL AUTONOMY & THE FINAL CHECKOFF

Focus: Liability mapping, EU AI Act, informed consent, independent operation. (*5-week month to absorb the winter holiday period.*)

Week 13 (Dec 24-30): Liability Mapping

Virtual (3 hrs)

- Who bears responsibility when an AI recommendation is wrong? Clinician vs. vendor vs. institution
- Practice-based learning and continuous professional development obligations
Hospital Practice (3 hrs)
- Case conference: real (de-identified) adverse-event report involving AI-assisted decision-making **Key References**
- NIH/AAMC liability & practice-based learning — pmc.ncbi.nlm.nih.gov/articles/PMC11993958 **Discussion:** Q4.3 (liability attribution)

Week 14 (Dec 31-Jan 6): EU AI Act Risk Tiers

Virtual (3 hrs)

- Risk-tier structure of the EU AI Act (unacceptable / high / limited / minimal risk)
- Mapping course tools (scribes, CDSS, care-planning NLP) to the correct tier **Hospital Practice (3 hrs)**
- Tier-classification exercise using the hospital's actual deployed AI tools **Key References**
- EU AI Act overview — [germancomplianceinstitute.com](https://www.germancomplianceinstitute.com) **Discussion:** Q4.4 (risk-tier matching)

Week 15 (Jan 7-13): Informed Consent & Transparency

Virtual (3 hrs)

- Patient right to know AI was involved; how to disclose without undermining trust
- Scripted consent-conversation practice **Hospital Practice (3 hrs)**
- Standardized-patient encounters requiring AI-use disclosure **Key References**
- EU AI Act & patient consent — pmc.ncbi.nlm.nih.gov/articles/PMC12973773 **Discussion:** Q4.1 (disclosure scenario)

Week 16 (Jan 14-20): Capstone Preparation

Virtual (3 hrs)

- Walkthrough of the “Solo Pilot” simulation structure and grading rubric
- Integration review: HITL (M1) + override (M2) + sovereignty (M3) + liability/consent (M4) **Hospital Practice (3 hrs)**
- Dry-run capstone shift (ungraded) with instructor feedback **Discussion:** Q4.2 (imaging discrepancy capstone-style case)

Week 17 (Jan 21-27): Month 4 Synthesis & Checkoff

Virtual (3 hrs)

- Final review session; open Q&A on liability, consent, and AI Act tiering **Hospital Practice (3 hrs)**
- Month 4 practical checkoff: mock disclosure conversation + risk-tier classification task, graded **Milestone:** Month 4 Competency Checkoff (Ethicolegal Autonomy) — pass/fail gate to Consolidation

MONTH 5 — CONSOLIDATION & MOCK OSCE

Focus: Cross-chapter integration; identifying compounded failures.

Week 18 (Jan 28-Feb 3): Integration Station Design

Virtual (3 hrs)

- Build integrative cases combining scribing error + diagnostic override + sovereignty documentation + consent disclosure **Hospital Practice (3 hrs)**
- Students design and pilot one integration station on a peer group

Week 19 (Feb 4-10): Mock OSCE Round 1

Virtual (3 hrs) — Debrief format explained **Hospital Practice (3 hrs)** — Timed OSCE stations, instructor-scored **Discussion:** Q5.1 (full HITL checkpoint walkthrough)

Week 20 (Feb 11-17): Mock OSCE Round 2 + Remediation

Virtual (3 hrs) — Individual feedback sessions **Hospital Practice (3 hrs)** — Repeat stations for students needing remediation **Discussion:** Q5.2 (automation bias vs. explainability failure)

Week 21 (Feb 18-24): Consolidation Wrap-Up

Virtual (3 hrs) — Full-class synthesis discussion; open forum on lingering uncertainties **Hospital Practice (3 hrs)** — Free-practice sandbox time, student-directed **Milestone:** Consolidation Sign-Off — readiness gate to Capstone

■ MONTH 6 — SOLO PILOT CAPSTONE & CERTIFICATION

Focus: Full independent clinical shift simulation; certification.

Week 22 (Feb 25-Mar 3): Capstone Shift 1

Full simulated shift: intake (AI scribing) → diagnosis (AI imaging validation) → treatment planning, graded against the Month 4 “Confidence Factor” checklist.

Week 23 (Mar 4-10): Capstone Shift 2 + Individual Debrief

Second graded shift with a different case mix; one-on-one debrief with faculty.

Week 24 (Mar 11-17): Remediation / Retake Window

For any student not yet at checkoff standard; open sandbox access for final practice.

Week 25 (Mar 18-24): Certification Ceremony & Course Close

Virtual (3 hrs) — Final exam review, course evaluation **Hospital Practice (3 hrs)** — Final

checkoff, AI Competency Certificate awarded **Milestone:** AI Competency Certificate issued (NIH/AAMC + German/EU dual-compliant)

FULL REFERENCE LIBRARY (numbered as in original blueprint, mapped to weeks)

#	Source	Topic	Used in Week(s)
1	pubmed.ncbi.nlm.nih.gov/36731054	NIH/AAMC Clinical AI Competency Model	1
2	germancomplianceinstitute.com — AI in Healthcare Germany: Legal, Ethical, GDPR Compliance Guide 2026	EU AI Act & GDPR	10, 14
3	pmc.ncbi.nlm.nih.gov/articles/PMC11993958	Clinical AI evidence base, bias case studies, liability	1, 3, 7, 13
4	youtube.com/watch?v=RkefjoU7qkM	German quality pillars overview (video)	6, 9
5	pmc.ncbi.nlm.nih.gov/articles/PMC11822726	Prompt literacy in clinical training	2
6	pubmed.ncbi.nlm.nih.gov/42710030	Human-in-the-Loop workflow	1
7	pmc.ncbi.nlm.nih.gov/articles/PMC12820482	Clinical AI evaluation methods	5
8	aafp.org/fpm/2025/0900/effective-ai-prompts	AAFP “Four Rs” prompting framework	2, 4
9	pmc.ncbi.nlm.nih.gov/articles/PMC12978887	GDPR / Explainable AI (XAI)	6
10	physicianaihandbook.com/future/medical-education.html	German quality pillars (Functionality/Robustness/Comprehensibility)	6
11	vde.com/de/dgbmt event proceedings	Local/open-weight deployment rationale	10
12	news.syenza.com/ai-models-german-medical-licensing-exams	AI board-exam vs. bedside reasoning dissociation	5
13	pmc.ncbi.nlm.nih.gov/articles/PMC7980122	AI-enhanced clinical encounters / care planning	9, 13
14	pubmed.ncbi.nlm.nih.gov/35612209	Final exam / capstone checklist methodology	16, 22–25
15	pmc.ncbi.nlm.nih.gov/articles/PMC12973773	DiGA (Digital Health Applications) pathway	11, 15

Note: instructors should independently verify each link’s current content before distribution, since URLs and article availability can change over time.

USMLE-STYLE DISCUSSION QUESTION BANK (full text, organized by week)

Month 1 — Foundation

Q1.1 (Week 1) — A 34-year-old family physician uses an ambient AI scribe during a visit for a patient with chest pain radiating to the jaw. The AI-generated note documents “patient denies chest pain” based on a mistranscription. The physician signs the note without reading it and the patient later returns with an MI. What is the *most important* systemic

failure illustrated here? A) The scribe's speech-recognition accuracy B) Automation bias / failure of human-in-the-loop verification C) Inadequate malpractice insurance D) Lack of ICD-10 coding accuracy E) Patient's own recall error **Answer: B.** HITL means the human must actively verify AI output, not passively co-sign it.

Q1.2 (Week 2) — Which prompt best exemplifies the AAFP "Four Rs" framework (Request, Reason, References, Revise)? A) "Summarize this." B) "Summarize this visit for chronic pain follow-up, cite which patient statements support each assessment, and flag anything uncertain for me to revise." C) "Write a note that will pass insurance audit." D) "Make this note sound more professional." E) "Translate this transcript into German." **Answer: B.** Only B covers Request, Reason, References, and Revise together.

Q1.3 (Week 3) — A German GP with limited technical background is onboarded to an ambient scribe. What is the primary regulatory driver for structured (not self-study) training? A) EU trade tariffs on software B) The documented shortage of technical AI literacy among German primary care physicians C) US FDA device classification D) HIPAA training requirements E) ICD-11 rollout timelines **Answer: B.**

Q1.4 (Week 4) — A resident says, "The scribe saved me two hours today, so I trust it completely now." What teaching point applies? A) Time saved is inversely proportional to error rate B) Efficiency gains do not change the verification obligation — time savings and output accuracy are separate axes C) The resident should stop using the tool D) Only attendings should use ambient scribes E) Time savings validate the tool's diagnostic accuracy **Answer: B.**

Month 2 — Diagnostics & Override

Q2.1 (Week 7) — A radiology CDSS flags a chest CT as "no acute findings" with 98% confidence, but the resident spots a subtle nodule. Appropriate action? A) Trust the 98% score and sign out B) Override the AI output, document the discrepancy, and route for independent radiologist review C) Increase the AI's confidence threshold setting D) Ignore the finding since AI systems are pre-validated E) Re-run the same model on the same image **Answer: B.**

Q2.2 (Week 6) — A hospital CDSS gives a recommendation but cannot show which variables drove it. Which German quality pillar is violated? A) Functionality B) Robustness C) Comprehensibility (Explainable AI) D) Interoperability E) Scalability **Answer: C.**

Q2.3 (Week 7) — Instructors seed a biased case producing a systematically wrong risk score. What behavior should be rewarded? A) Accepting the score because it matches protocol B) Detecting the anomaly, articulating clinical implausibility, and executing a manual override C) Reporting the vendor immediately without internal review D) Disabling the CDSS permanently E) Waiting for IT before treating the patient **Answer: B.**

Q2.4 (Week 5) — An LLM answers board questions correctly at high rates but fails multi-step bedside reasoning. This illustrates: A) The AI is unsafe for any clinical use B) Test-taking performance does not guarantee real-world clinical reasoning competence C) USMLE questions are poorly designed D) AI models cannot process medical vocabulary E) Board scores are the best proxy for bedside safety **Answer: B.**

Month 3 — Care Pathways & Sovereignty

Q3.1 (Week 10) — A German hospital is blocked from using a US cloud LLM API for nursing care plans. Most likely GDPR reason? A) The vendor's model is outdated B) Cross-border transfer of identifiable patient data without adequate safeguards C) Insufficient bandwidth D) US models can't process German E) API lacks HL7 support **Answer: B.**

Q3.2 (Week 9) — A nurse uses AI to draft a care plan combining NLP extraction with a

standardized nursing model. What must happen before finalizing? A) Auto-publish to EHR B) Clinician review and modification against actual patient presentation C) Skip review since NLP is deterministic D) Vendor approval E) Delay care for manual re-typing **Answer: B.**

Q3.3 (Week 11) — A researcher wants to use AI-extracted hospital data for a retrospective study. What must be true first? A) Data must stay fully identifiable for accuracy B) Anonymized/pseudonymized per GDPR data-minimization principles C) Verbal consent from the department head suffices D) Stored on the researcher's personal laptop E) No special handling needed **Answer: B.**

Q3.4 (Week 10) — Why prefer an on-premises open-weight LLM over a proprietary cloud model in an EU hospital EHR? A) Open-weight models are always more accurate B) On-premises deployment keeps data within jurisdictional control, avoiding cross-border issues C) Cloud models can't be updated D) Open-weight models need no maintenance E) Proprietary models are banned globally **Answer: B.**

Month 4 — Ethicolegal Autonomy

Q4.1 (Week 15) — A patient asks whether AI was involved in generating their diagnostic report. Appropriate response? A) Deny AI involvement B) Disclose AI-assisted tools were used, explain clinician verification role, answer questions C) Refuse — proprietary D) Redirect to the vendor only E) Say nothing unless formally requested **Answer: B.**

Q4.2 (Week 16) — During the capstone, the AI imaging tool disagrees with the student's own fracture read. Best demonstration of competency? A) Automatically defer to the AI B) Independently re-examine the image, reason through the discrepancy, document the final decision with justification C) Order a repeat scan to avoid deciding D) Ask a classmate to decide E) Accept whichever answer came first **Answer: B.**

Q4.3 (Week 13) — If a clinician follows a wrong AI recommendation, who bears primary clinical responsibility under EU/German liability frameworks? A) The AI vendor exclusively B) No one — AI decisions are autonomous C) The treating clinician, who retains independent judgment duty D) The hospital IT department E) The patient, for consenting to AI-assisted care **Answer: C.**

Q4.4 (Week 14) — Which correctly matches an EU AI Act risk tier to a clinical AI use case? A) Ambient scribing = prohibited practice B) Diagnostic CDSS influencing treatment = high-risk system requiring oversight, transparency, robustness C) Hospital Wi-Fi = high-risk AI system D) Care-plan NLP tools = banned under the AI Act E) All clinical AI is classified identically **Answer: B.**

Consolidation

Q5.1 (Week 19) — Mock OSCE: an ambient-scribed, AI-flagged "low risk" chest pain note for a patient with new hypotension. Walk through every HITL checkpoint from Months 1–4 before disposition. *Rubric: verify transcription accuracy → question the AI risk score against exam findings → invoke manual override (M2) → confirm anonymization/documentation compliance (M3) → document independent reasoning and consent disclosure (M4).*

Q5.2 (Week 20) — Compare "automation bias" (M1) with "explainability failure" (M2) using one clinical example. *Discussion: automation bias is a human failure to scrutinize; explainability failure is a system failure to justify. Strongest examples show both compounding — an opaque output accepted without scrutiny.*

Instructor Notes

- Pass/fail milestone checkoffs occur at the end of Months 1–4; students who fail a checkoff repeat that block’s hospital-practice hours before advancing.
- All open-access reference links should be re-verified each term, since article URLs and hosting can change.
- This package is designed to run as-is for an Oct 2026–Mar 2027 cohort; shift the calendar forward by exactly the same weekday offsets for future cohorts.